THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · MEDIUM LEVEL UNIT TEST SET 1

Chapters 9–12: **Light - Reflection and Refraction, The Human Eye and the Colourful World, Electricity, Magnetic Effects of Electric Current** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. An object is placed at the centre of curvature of a concave mirror of focal length 12 cm. The distance of the image from the mirror is 1

(a) 24 cm **(b)** 12 cm **(c)** 36 cm **(d)** 6 cm

2. A person with myopia has a far point at 100 cm. The lens needed has a power of 1

(a) +10 D **(b)** -0.1 D **(c)** -1 D **(d)** +1 D

3. The number of electrons passing per second through a conductor carrying 3.2 A is 1

(a) 2 x 10^18 **(b)** 3.2 x 10^19 **(c)** 5 x 10^18 **(d)** 2 x 10^19

4. A compass needle placed below a wire carrying current from south to north is deflected. If the current is reversed, the deflection 1

(a) stays the same **(b)** is reversed **(c)** doubles **(d)** becomes zero

5. If n = sin i / sin r for glass is 1.5 and sin i = 0.75, then sin r is 1

(a) 0.25 **(b)** 0.75 **(c)** 0.5 **(d)** 1.125

6. A beam of white light is passed through a prism and a screen shows 7 colours. Replacing white by red light gives 1

(a) no light **(b)** 7 colours
(c) a single red patch **(d)** white light

7. Three resistors of 6 ohm each are connected in parallel. The equivalent resistance is 1

(a) 6 ohm **(b)** 18 ohm **(c)** 3 ohm **(d)** 2 ohm

8. A straight conductor carrying current from east to west is placed in a magnetic field pointing down. The force is directed 1

(a) downward **(b)** to the south **(c)** upward **(d)** to the north

9. An object is placed 40 cm from a convex lens of f = 20 cm. The image distance is 1

(a) 60 cm **(b)** 20 cm **(c)** 10 cm **(d)** 40 cm

10. If the atmosphere of the earth were absent, the day sky would appear 1

(a) red **(b)** blue **(c)** white **(d)** dark

11. The power of an electric device that draws 5 A at 220 V is 1

(a) 44 W **(b)** 225 W **(c)** 1.1 W **(d)** 1100 W

12. If a magnet is moved faster into a coil, the induced current 1

(a) becomes zero **(b)** decreases **(c)** increases **(d)** remains the same

13. A concave mirror of f = 15 cm forms an image of the same size as the object. The object distance is 1

(a) 30 cm **(b)** 7.5 cm **(c)** 15 cm **(d)** 45 cm

14. A person's near point is 50 cm. To read at 25 cm, he needs a lens of power 1

(a) +4 D **(b)** +2 D **(c)** -2 D **(d)** +0.5 D

15. A charge of 600 C flows in 5 minutes. The current is 1

(a) 120 A **(b)** 0.5 A **(c)** 2 A **(d)** 3 A

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. An object is placed 30 cm from a concave mirror. The image is formed at 30 cm on the same side. Find the focal length. 2

17. A person has a near point at 40 cm. Find the focal length of the lens needed to read at 25 cm. 2

18. How many electrons flow per second through a conductor carrying a current of 0.8 A? 2

19. State why two field lines cannot intersect each other. 2

20. The refractive index of a medium is 1.6. Find the speed of light in that medium. 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. An object 6 cm high is placed 20 cm in front of a concave mirror of f = 15 cm. Find the image distance, size and nature. 3

22. A person cannot see beyond 1.5 m clearly. Find the focal length, power and nature of the corrective lens. 3

23. A battery of 9 V is connected to 3 ohm, 4 ohm and 2 ohm in series. Find the current and the voltage across the 4 ohm resistor. 3

24. A bar magnet is moved towards a coil connected to a galvanometer and then away. Describe the observations and give the reason. 3

OR

A concave lens of f = -10 cm forms an image of an object at 20 cm. Find the image distance, nature and magnification. 3

25. Explain why the colour of the Sun is red during sunset but white at noon. 3

OR

Explain with the help of a formula why a thick wire is used for heavy current. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. An object 3 cm high is placed 20 cm from a concave mirror of focal length 12 cm. Find the position, nature, size of the image and magnification. Draw the ray-diagram rules used. 5

OR

A person has a near point of 1 m and a far point at infinity. 5

(a) Name the defect

(b) find the lens power to read at 25 cm

(c) state the nature of the lens.

27. A circuit has three resistors: 6 ohm and 3 ohm in parallel, in series with 4 ohm, connected to a 12 V battery. Find 5

(a) the total resistance

(b) the current from the battery

(c) the voltage across the parallel combination

(d) the current through each parallel resistor.

OR

Explain the magnetic field pattern around a current-carrying straight wire and how it changes when the current is doubled and when the distance from the wire is doubled. Describe how to show the direction of the field using the right-hand thumb rule. 5

END OF QUESTION PAPER **Total: 50 Marks**

MASTERMIND